Web Science

Assignment – 1

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1.Cable Issue:

Let us consider two cables of 20 meters each. One of them is in a 100MBps network while the other is in a 10MBps network. If you had to transfer data through each of them, how much time would it take for the first bit to arrive in each setting? (For your calculation you can assume that the speed of light takes the same value as in the videos.) Please provide formulas and calculations along with your results.

Answer:

1. We have a 100MBit/s network. It means that we can send 100 million bits per second. So, for sending 1 bit we need 10 nanoseconds of time.

The data spreads at the speed of light (the speed of light is approximately 300 million m/s). So, we can multiply (speed of light * 10 ns). And what we see is 100MBit/s Ethernet device within 1 clock cycle the signal would have traveled 3 meters.

So, now applying speed distance time formula, we get

$$\text{time} = \frac{20\,m}{3\,m} * 10 \text{ ns}$$

time = 66.67 ns

Hence, it would take 66.67 ns for the first bit to arrive in the first setting with 100MBps Network and 20-meter cable.

2. We have a 10MBit/s network. It means that we can send 10 million bits per second. So, for sending 1 bit we need 100 nanoseconds of time.

The data spreads at the speed of light (the speed of light is approximately 300 million m/s). So, we can multiply (speed of light * 100 ns). And what we see is 10MBit/s Ethernet device within 1 clock cycle the signal would have traveled 30 meters.

So, now applying speed distance time formula, we get

$$\text{time} = \frac{20\,m}{30\,m} * 100 \text{ ns}$$

time = 66.67 ns

Hence, it would take 66.67 ns for the first bit to arrive in the setting with 10MBps Network and 20-meter cable.

3. Collision:

1. **Explain why collision occurs when sending an Ethernet frame.**

When node tries to transmit a frame without sensing the status of the network, it will lead to a collision with other nodes, if other nodes also send frames to the network at the same time.

For Example, let's assume a network with three nodes A, B and C. Now, A tries to send an ethernet frame to B and starts transmitting. At the same time, C also tries to send a frame to B. There will be collision of frames from A and C and B will receive corrupted Data

2. **Elaborate in steps how the collision detection algorithm works.**

The concept of Collision Detection Algorithm is explained below in a stepwise manner:

Step 1 – Assemble a frame. Wait for a frame, if it is not ready for transmission.

Step 2 – Transmit if network is not busy, else wait.

Step 3 – Start transmitting and monitor for collision during transmission.

Step 4 – If there is no collision, transmit the next bit of the frame.

Step 5 – If collision occurs, continue transmission until maximum number of attempts is reached. If there is collision after maximum attempts, then abort the transmission.

Step 6 – Set a 'back-off' time using random number (usually [0,2n-1]) based on number of collisions and wait.

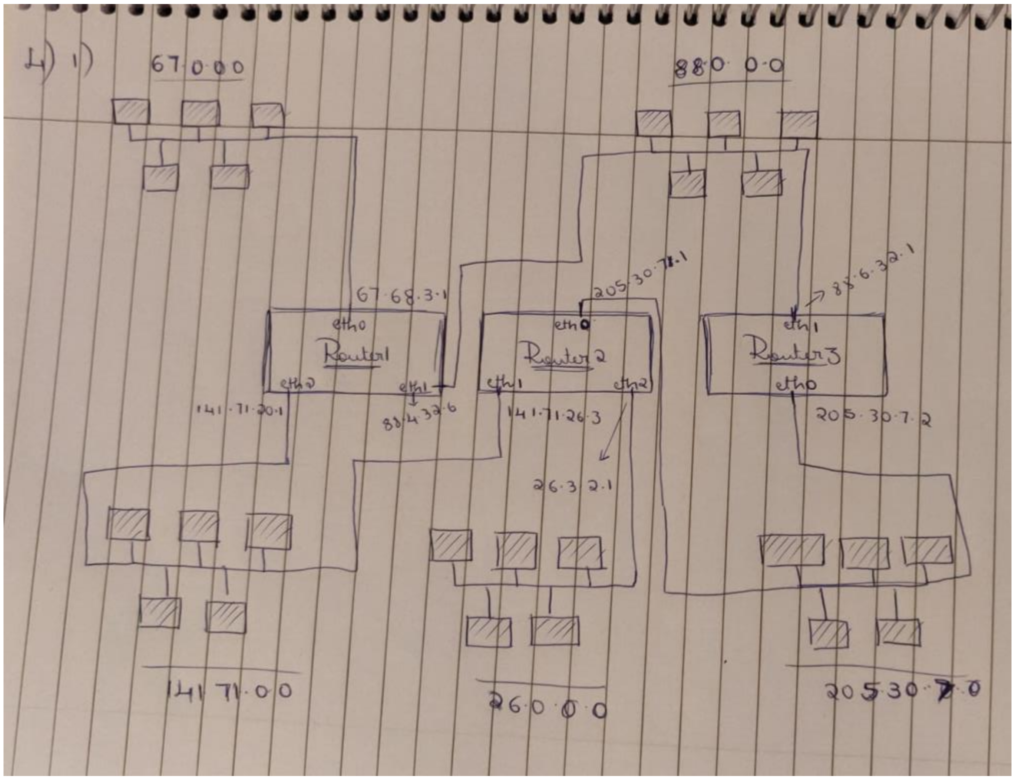
Step 7 – Attempt again to transmit from Step 1.

This procedure is used to initiate a transmission. It is complete when the frame is transmitted successfully, or a collision is detected during transmission

Reference: Wikipedia

4. Routing Table

1. **Draw the schematic representation of the network based on Routing Table.1**



2. **Find the shortest path of sending information from 205.30.7.0 network to 141.71.0.0 network.**

- The shortest path to send information from 205.30.7.0 network to 141.71.0.0 network is through Router 2.

-The routing table of the network 205.30.7.0 does not have a route to 141.71.0.0 in its routing table and hence takes the default next-hop 205.30.7.1, eth0.

- The routing table of Router 2 has the

Destination network -> 141.71.0.0

Next-hop -> 141.71.26.3 hence forwards the packet to its own eth1 device.

- The packet is forwarded to network 141.71.0.0